Gram–Schmidt Orthogonalization

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of gram–schmidt orthogonalization, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

LA-02 / Linear Algebra / Report v1

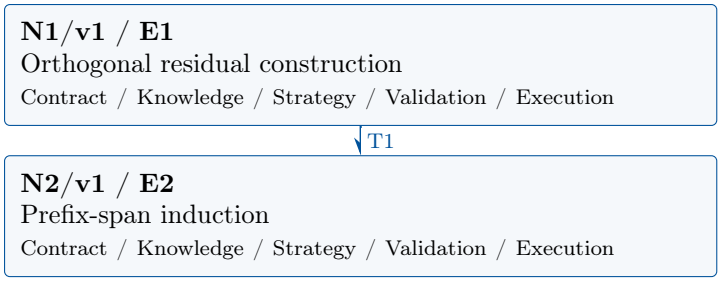
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Given linearly independent $v_1, \ldots, v_k$ in a real or complex inner-product space, prove Gram–Schmidt produces orthonormal $q_1, \ldots, q_k$ with equal prefix spans:* $\operatorname{span}(q_1, \ldots, q_j) = \operatorname{span}(v_1, \ldots, v_j)$.

A. Foundations and permitted imports

The inner product is linear in its first argument and conjugate-linear in its second, so projection onto unit q is $\langle v, q\rangle q$. Given vectors $v_1, \ldots, v_k$ are linearly independent. Use positivity of the norm, Pythagoras, and finite induction; the ambient inner-product space may be larger or infinite-dimensional.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. Each new q_j is defined and orthogonal to all previous q_i.

Receiving activity. N2: Prefix-span induction

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Residual formula and nonzero normalization; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Orthogonal residual construction

Object and role. Remove precisely the old-span component.

Inputs. Independent input vectors; existing orthonormal prefix; foundational imports stated in §1

C — Contract

Own E1/v1: Each new q_j is defined and orthogonal to all previous q_i.

K — Knowledge

Retain Independent input vectors; existing orthonormal prefix; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Remove precisely the old-span component.

V — Validation

Do not divide by a zero residual when input vectors are dependent.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Set $w_1 = v_1$ and $q_1 = w_1/\|w_1\|$.

N1.2 Suppose $q_1, \dots, q_{j-1}$ are orthonormal with the same prefix span as the first $j-1$ inputs.

N1.3 Define $w_j = v_j - \sum_{i<j} \langle v_j, q_i \rangle q_i$.

N1.4 For $\ell < j$, inner-product linearity gives $\langle w_j, q_\ell \rangle = \langle v_j, q_\ell \rangle - \langle v_j, q_\ell \rangle = 0$.

N1.5 If $w_j = 0$ then v_j is in the preceding input span, contradicting independence.

N1.6 Hence $q_j = w_j/\|w_j\|$ is defined and has unit norm.

Export E1/v1. Each new q_j is defined and orthogonal to all previous q_i.

Acceptance gate. Do not divide by a zero residual when input vectors are dependent. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Do not divide by a zero residual when input vectors are dependent. Recheck dependent nodes N2.

B. N2/v1: Prefix-span induction

Object and role. Prove both span inclusions at every prefix.

Inputs. Residual formula and nonzero normalization; prior exports E1/v1

C — Contract

Own E2/v1: An orthonormal list with all requested prefix-span equalities.

K — Knowledge

Retain Residual formula and nonzero normalization; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Prove both span inclusions at every prefix.

V — Validation

Only the span is canonical; signs/phases and input order can change the basis.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 The residual formula places q_j in span $(v_1, \ldots, v_j)$, since earlier q_i are in the preceding span.

N2.2 Conversely $v_j = \|w_j\| q_j + \sum_{i<j} \langle v_j, q_i \rangle q_i$, so v_j belongs to span $(q_1, \ldots, q_j)$.

N2.3 Earlier spans already coincide by induction.

N2.4 Thus both inclusions hold and the prefix spans agree.

N2.5 Induction completes the construction for all j; the empty input list gives an empty orthonormal list.

Export E2/v1. An orthonormal list with all requested prefix-span equalities.

Acceptance gate. Only the span is canonical; signs/phases and input order can change the basis. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Only the span is canonical; signs/phases and input order can change the basis. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Orthogonal residual construction

1. Set $w_1 = v_1$ and $q_1 = w_1/\|w_1\|$.
2. Suppose $q_1, \ldots, q_{j-1}$ are orthonormal with the same prefix span as the first $j-1$ inputs.
3. Define $w_j = v_j - \sum_{i<j} \langle v_j, q_i \rangle q_i$.
4. For $\ell < j$, inner-product linearity gives $\langle w_j, q_\ell \rangle = \langle v_j, q_\ell \rangle - \langle v_j, q_\ell \rangle = 0$.
5. If $w_j = 0$ then v_j is in the preceding input span, contradicting independence.
6. Hence $q_j = w_j/\|w_j\|$ is defined and has unit norm.

B. §4.N2 — Prefix-span induction

1. The residual formula places q_j in span $(v_1, \ldots, v_j)$, since earlier q_i are in the preceding span.
2. Conversely $v_j = \|w_j\| q_j + \sum_{i<j} \langle v_j, q_i \rangle q_i$, so v_j belongs to span $(q_1, \ldots, q_j)$.
3. Earlier spans already coincide by induction.
4. Thus both inclusions hold and the prefix spans agree.
5. Induction completes the construction for all j; the empty input list gives an empty orthonormal list.

□

V. WORKED ORCHESTRATION

For $v_1 = (1, 1)$, $v_2 = (1, 0)$ in R^2, $q_1 = (1, 1)/\sqrt{2}$. The second residual is $(1/2, -1/2)$, so $q_2 = (1, -1)/\sqrt{2}$. Both prefixes have the asserted spans, and the projection coefficient for v_2 is $1/\sqrt{2}$.

A. Boundary and hypothesis audit

For $v_2 = 2v_1$ the second residual is 0. Validation detects the failed independence condition before normalization. A modified algorithm may discard dependent vectors and compute a basis of the span, but that is a different output contract.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

For $v_2 = 2v_1$ the second residual is 0. Validation detects the failed independence condition before normalization. A modified algorithm may discard dependent vectors and compute a basis of the span, but that is a different output contract.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.